



HK IMO Training 2023-2024 Problem Set 1 Suggested Solutions



Problem 1 (Canada MO 2018 Problem 1). There are n points on the plane which are not necessarily distinct. In each step, we can move two points to their midpoint. Our goal is to move all n points to the same point after a finite number of moves. Prove that we can meet our goal regardless of the initial positions of the n points if and only if n is a power of 2.

Solution. If $n = 2^k$ for some nonnegative integer k , we prove that the goal can be met by induction on k . The base case $k = 0$ is trivial. For the case k , we first use the inductive hypothesis to move 2^{k-1} of the points to the same point A , then move the other 2^{k-1} points to the same point B after some moves. Finally, we move one point at A and one point at B to their midpoint every time. After 2^{k-1} more moves, all points are moved to the midpoint of AB as desired. This proves the result for $n = 2^k$.

If n is not a power of 2, we claim that the goal cannot be met if all points are put on the same line (say, the real number line), with $n - 1$ of them at 0 and the other at 1. Note that the points always stay on the real number line, and the sum of all numbers is always 1 no matter how we move the points. If all points are moved to the same point, then all of them must be moved to $\frac{1}{n}$ at the end. However, in each move we can only replace two numbers by their average, and so the denominator of any number formed must be a power of 2. Therefore, it is impossible that any of the numbers is moved to $\frac{1}{n}$. This means the goal cannot be met. $\square$

Problem 2 (Bay Area MO 2016 Grade 12 Problem 4). Does there exist a positive integer n such that

$$\pm 1^3 \pm 2^3 \pm 3^3 \pm \cdots \pm n^3 = 20162016$$

for some suitable choices of the signs?

Solution. Yes. Using finite differences, we find that

$$\begin{aligned}(x+1)^3 - x^3 &= 3x^2 + 3x + 1, \\ ((x+3)^3 - (x+2)^3) - ((x+1)^3 - x^3) &= (3(x+2)^2 + 3(x+2) + 1) - (3x^2 + 3x + 1) = 12x + 18,\end{aligned}$$

and

$$\begin{aligned}&((x+7)^3 - (x+6)^3 - (x+5)^3 + (x+4)^3) - ((x+3)^3 - (x+2)^3 - (x+1)^3 + x^3) \\&= (12(x+4) + 18) - (12x + 18) \\&= 48.\end{aligned}$$

In other words, we can choose the signs of any 8 consecutive terms in an appropriate way such that the sum is equal to 48. As $20162016 = 48 \times 420042$, we can make up the number 20162016 by using $n = 8 \times 420042$ terms. $\square$

Problem 3 (Bay Area MO 2019 Grade 12 Problem 4). Let S be a finite set of nonzero real numbers, and let $f : S \rightarrow S$ be a function. Suppose for every $x \in S$, we have $f(f(x)) = x + f(x)$ or $f(f(x)) = \frac{x + f(x)}{2}$. Prove that $f(x) = x$ for all $x \in S$.

Solution. We first consider some $a_1 \in S$ with $f(a_1) \neq a_1$ such that $f^n(a_1) = a_1$ for some positive integer $n \geq 2$, where n is chosen to be the smallest such integer. Let $a_{j+1} = f(a_j)$ for each $j \geq 1$.

If there are two consecutive terms a_j and a_{j+1} which are both positive, then a_{j+2} is also positive because it is equal to $a_j + a_{j+1}$ or $\frac{a_j + a_{j+1}}{2}$ by the given condition. Inductively, it is easy to see that every term in the sequence is positive (including $a_1, \dots, a_{j-1}$

because the terms are periodic). There exist $b_3, b_4, \dots, b_{n+2} \in \left\{1, \frac{1}{2}\right\}$ such that

$$\begin{aligned}a_3 + a_4 + \cdots + a_{n+2} &= b_3(a_1 + a_2) + b_4(a_2 + a_3) + \cdots + b_{n+2}(a_n + a_{n+1}) \\&\geq \frac{a_1 + a_2}{2} + \frac{a_2 + a_3}{2} + \cdots + \frac{a_n + a_{n+1}}{2} \\&= \frac{1}{2}a_1 + a_2 + \cdots + a_n + \frac{1}{2}a_{n+1}.\end{aligned}$$

But since $a_{n+1} = a_1$ and $a_{n+2} = a_2$, the two sides are equal. In order that the equality holds, we must have $a_{j+2} = \frac{a_j + a_{j+1}}{2}$ for every j . But this cannot happen when a_{j+2} is the largest term in the sequence (which exists as the sequence is periodic), since $a_j \leq a_{j+2}$ and $a_{j+1} < a_{j+2}$.

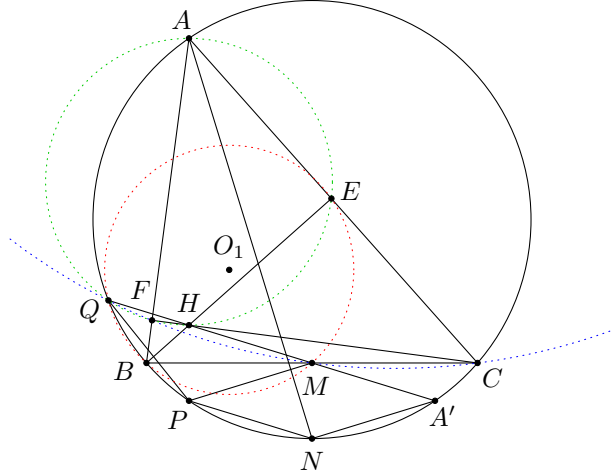
Similarly, if there are two consecutive terms which are both negative, we can derive a contradiction. Therefore, the signs of such a sequence must be alternating. WLOG assume a_1 is the largest term in the sequence (hence $a_1 > 0$ and $a_2 < 0$). In order that $a_3 > 0$, we must have $a_1 + a_2 > 0$. This implies $|a_1| > |a_2|$. Similarly, in order that $a_4 < 0$, we must have $a_2 + a_3 < 0$, and hence $|a_2| > |a_3|$. Inductively, we find that $|a_j| > |a_{j+1}|$ for all j , which is impossible as the sequence is periodic.

We have shown that such a sequence cannot exist. Now, for any $x \in S$, there exist integers $m > k \geq 0$ such that $f^k(x) = f^m(x)$ since S is finite. We choose the smallest such k and then the smallest such m . If $k = 0$, i.e. $x = f^m(x)$, then we must have $m = 1$ from above. This means $f(x) = x$ as desired. If $k \geq 1$, again we must have $m = k + 1$ from above by considering $a_1 = f^k(x)$. Let $a = f^{k-1}(x)$. Then $f^k(x) = f^{k+1}(x)$ means $f(a) = f(f(a))$.

- If $f(f(a)) = a + f(a)$, then $a = 0$, which contradicts the given condition.
- If $f(f(a)) = \frac{a + f(a)}{2}$, then $f(a) = a$. This implies $f^k(x) = f^{k-1}(x)$, which contradicts the minimality of k .

To sum up, the only possibility is $f(x) = x$ for all $x \in S$. □

Problem 4 (USA TST 2023 Problem 2). In an acute $\triangle ABC$, let E and F be the feet of altitudes from B and C respectively, and let M be the midpoint of BC . The common external tangents to the circumcircles of $\triangle BEM$ and $\triangle CFM$ intersect at a point P . If P lies on the circumcircle of $\triangle ABC$, prove that $AP \perp BC$.



Solution. Let O_1 and O_2 be the centres of (BME) and (CMF) respectively, and let Q be the other intersection point of these two circles.

Firstly, it is well-known that $MB = ME$. Therefore, O_1M is the perpendicular bisector of BE . In particular, $O_1M \perp BE$, and hence $O_1M \parallel AC$. Similarly, $O_2M \parallel AB$. It follows that $\angle O_1MO_2 = \angle BAC$. By homothety, we have $\frac{PO_1}{PO_2} = \frac{MO_1}{MO_2}$. Therefore, PM is the external angle bisector of $\angle O_1MO_2$, which is parallel to the external angle bisector of $\angle BAC$. Let N be the midpoint of the minor arc BC of (ABC) . As AN is the internal angle bisector of $\angle BAC$, we have $AN \perp PM$. Let A' be the point on (ABC) such that AA' is a diameter. Then $PM \parallel NA'$ because $\angle ANA' = 90^\circ$.

Secondly, using directed angles, we deduce

$$\angle BQC = \angle BQM + \angle MQC = \angle BEM + \angle MFC = \angle MBE + \angle FCM = \frac{\pi}{2} - \angle ACB + \frac{\pi}{2} - \angle CBA = \angle BAC$$

and

$$\angle FQE = \angle FQM + \angle MQE = \angle FCM + \angle MBE = \frac{\pi}{2} - \angle CBA + \frac{\pi}{2} - \angle ACB = \angle FAE.$$

These show Q lies on both (ABC) and (AEF) . It is well-known that Q, H, M, A' are collinear where H is the orthocentre of $\triangle ABC$. As P lies on O_1O_2 , which is the perpendicular bisector of MQ , we know that $PM = PQ$.

As $\angle QA'N = \angle QMP = \angle PQM$, the cyclic quadrilateral $PNA'Q$ is an isosceles trapezoid with $PN \parallel QA'$. Therefore, $MPNA'$ is a parallelogram. Note that the circumcentre of $\triangle PNA'$ lies on the line MN , because MN is the perpendicular bisector of BC . This is only possible when $MPNA'$ is a rhombus (since otherwise the perpendicular bisector of PA' only intersects the line MN at the midpoint of MN). This means P and A' are symmetric about the line MN , and hence $CBPA'$ is an isosceles trapezoid. It follows that $AP \perp BC$. □